SHAHRIAR TASNIM

Electrical Engineer | Circuit & PCB Design | Embedded Hardware | Test & Validation | Open to relocation
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SUMMARY

Electrical engineer who designs, builds, and validates electronic hardware. Took microcontroller PCBs from schematic capture to fabricated board: an ESP8266 IoT sensor node and a 43A H-bridge motor-control board for a University Rover Challenge finalist rover. Backed by 2 years of hardware testing, fault isolation, and validation with oscilloscopes and DMMs, with clear test documentation and close work with vendors and multidisciplinary teams.

TECHNICAL SKILLS

- **Electrical design tools:** Altium Designer (schematic capture, PCB layout, Gerber/drill outputs, BOM); Multisim, Proteus; MATLAB
- **Circuit & hardware design:** Microcontroller circuits (ESP8266, ATmega2560), voltage regulation and power management, high-current H-bridge motor drivers (BTS7960, 43 A), analog/capacitive sensor acquisition, I^2C peripherals, through-hole/SMD boards
- **Test & validation:** Oscilloscope, DMM, clamp meter; functional and integration testing, fault isolation, calibration
- **Embedded & software:** Embedded C, C++, Python, ROS 2, Linux, Git, MQTT | **Safety:** NFPA 70E, LOTO

DESIGN PROJECTS

IoT Soil Moisture Monitoring Device: Custom PCB

- Designed the schematic and PCB layout in Altium Designer for an ESP8266 wireless sensor node with capacitive moisture sensing, I^2C OLED display, USB power/programming, voltage regulation, and expansion GPIO; produced Gerber, drill, and BOM outputs for fabrication.
- Wrote C++ firmware for sensing, alerts, MQTT/ThingSpeak cloud upload, and deep-sleep low-power operation.

University Rover Challenge Finalist: Rover Control Unit PCB **May 2019**
Mars Desert Research Station (MDRS) *Utah, USA*

- Selected as one of 36 finalists from 136 teams worldwide; placed 18th, contributing to electrical, embedded, and robotics systems for autonomous and teleoperated rover operation.
- Designed and fabricated a custom control PCB around an Arduino Mega (ATmega2560), interfacing a BTS7960 43A full H-bridge for high-current DC motor control and a soil pH sensor for on-board data acquisition.

EXPERIENCE

Graduate Assistant, Robotics & Intelligent Control Systems Lab **Jan 2025 – Apr 2026**
Lamar University *Beaumont, TX*

- Integrated and validated a robotic cell (xArm5, RealSense, ROS 2) via bring-up, calibration, and 300 structured trials.
- Isolated intermittent sensing, motion, I/O, and communication faults; validated fixes and documented test procedures.
- Reduced average cycle time 34.6% (63.0 s $\rightarrow$ 41.2 s) through systematic testing and fault isolation.

Electrical Engineer **Oct 2021 – Apr 2023**
STATA IT Limited *Dhaka, Bangladesh*

- Diagnosed and repaired 100+ faulty IoT devices with the R&D team using schematics, oscilloscopes, and DMMs; reported recurring failures to vendors with root-cause findings and proposed fixes.
- Installed, configured, and functionally tested 250+ smart-home devices at customer sites.
- Led site visits, product demos, technical support, and quotes, translating customer requirements into system solutions.

Electrical Engineering Intern **Mar 2021 – Sep 2021**
A A Rashayan Shilpa Limited *Dhaka, Bangladesh*

- Supported testing and commissioning of motors, transformers (up to 480 V), breakers, and power distribution.

EDUCATION & CERTIFICATIONS

Master of Engineering Science in Electrical Engineering **May 2026**
Lamar University, Beaumont, TX GPA: 3.67/4.00

Bachelor of Science in Electrical & Electronic Engineering **Oct 2021**
American International University–Bangladesh, Dhaka, Bangladesh GPA: 3.38/4.00

Relevant coursework: VLSI Circuits, Analog Electronics I & II, Electronic Devices, Microprocessors & Embedded Systems, Digital Signal Processing, Industrial Electronics & Drives

Certifications: NFPA 70E (OSHA, 2026) • Industrial Sensors for PLCs • AIoT: Connected Products (NASBA, 2026)